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ELECTRONIC PRINCIPLES & CIRCUITS

BEC303 – MODULE 3

Linear Op-Amp Circuits | Comparators | Oscillators | 555 Timer

1. LINEAR OP-AMP CIRCUITS

1.1 Summing Amplifier

A summing amplifier combines multiple input voltages into a single weighted output. It uses the inverting op-amp configuration with multiple input resistors connected to the virtual ground at the inverting terminal.

Working Principle: Due to the virtual short, the inverting terminal is at 0V. Each input current ($I_1=V_1/R_1$, $I_2=V_2/R_2$, etc.) flows through its resistor to the summing node. By KCL, the total current flows through R_f , generating the output voltage.

Output Formula: $V_{out} = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$

If all input resistors are equal ($R_1=R_2=R_3=R$): $V_{out} = -(R_f/R)(V_1+V_2+V_3)$

Applications: Audio mixers, DAC circuits, analog computing, signal averaging.

1.2 The Subtractor

A subtractor produces an output equal to the difference of two input voltages. It uses two op-amp stages: the first stage inverts V_1 (gain = -1), producing $-V_1$. The second stage is a summing amplifier that adds $-V_1$ and V_2 .

Output: $V_{out} = V_2 - V_1$

Single op-amp difference amplifier (with R on all terminals): $V_{out} = (R_f/R_1)(V_2 - V_1)$

Note: All resistors must be precisely matched for accurate subtraction. Mismatched resistors introduce common-mode error.

1.3 The Averager

An averager is a summing amplifier that produces the arithmetic average of all input voltages. Each input resistor is set to nR (where n = number of inputs) and the feedback resistor = R .

Each channel gain = $R/nR = 1/n$

Output: $V_{out} = -(V_1 + V_2 + \dots + V_n) / n$

Any number of inputs can be averaged by setting all input resistors to nR and feedback resistor to R .

1.4 D/A Converter – Binary Weighted

A Digital-to-Analog (D/A) converter takes a binary digital input and produces a proportional analog voltage. The binary-weighted DAC uses the summing amplifier concept with binary-weighted input resistors: R , $2R$, $4R$, $8R$ (for a 4-bit converter).

Channel gains for 4-bit converter:

MSB (v_3): $A_v = -1$ v_2 : $A_v = -0.5$ v_1 : $A_v = -0.25$ LSB
(v_0): $A_v = -0.125$

Output: $V_{out} = -(v_3 + 0.5v_2 + 0.25v_1 + 0.125v_0)$

With 4 bits, 16 combinations are possible (0000 to 1111). When all inputs are 1111: $V_{out} = -(1+0.5+0.25+0.125) = -1.875V$ (maximum). The output looks like a negative staircase on an oscilloscope.

Disadvantage: Requires a wide range of precisely matched resistors (R to $2^{(n-1)}R$). For 12-bit, ratio = 2048:1 — difficult to manufacture accurately.

1.5 R/2R Ladder D/A Converter

The R/2R ladder DAC solves the precision resistor problem by using ONLY TWO resistor values: R and $2R$, regardless of the number of bits. Binary switches D_0 – D_3 connect $2R$ resistors to either ground (logic 0) or $+V_{ref}$ (logic 1).

Key property: Looking into any node, Thevenin equivalent resistance is always R . Each bit contributes exactly half the weight of the previous bit.

BIN = $(D_0 \times 2^0) + (D_1 \times 2^1) + (D_2 \times 2^2) + (D_3 \times 2^3)$

$V_{out} = -(BIN / 2^N) \times 2V_{ref}$

Example: $D_0=1$, $D_1=0$, $D_2=0$, $D_3=1$, $V_{ref}=+5V \rightarrow BIN=9 \rightarrow V_{out} = -(9/16) \times 10 = -5.625V$

Advantages over binary-weighted: Only 2 resistor values, better accuracy, easily scalable, widely used in ICs.

2. NONLINEAR OP-AMP CIRCUITS

2.1 Comparator with Zero Reference

A comparator uses an op-amp without feedback, operating in open-loop. Due to very high open-loop gain ($AVOL \sim 100,000$), even a tiny input drives the output to saturation.

Non-Inverting Comparator:

Input V_{in} applied to (+) terminal. (–) terminal grounded. When $V_{in} > 0 \rightarrow \text{output} = +V_{sat}$. When $V_{in} < 0 \rightarrow \text{output} = -V_{sat}$. Called a zero-crossing detector. Converts sine wave to square wave.

Minimum input to saturate: $V_{in(min)} = \pm V_{sat} / AVOL = \pm 14 / 100000 = \pm 0.14 \text{ mV}$

Inverting Comparator:

Input applied to (–) terminal, (+) grounded. Behavior is OPPOSITE: $V_{in} > 0 \rightarrow -V_{sat}$; $V_{in} < 0 \rightarrow +V_{sat}$. Diode clamps at input protect against excessively large voltages (maintains $\pm 0.7 \text{ V}$ at op-amp input).

2.2 Comparator with Non-Zero Reference

Used when detection at a non-zero threshold is needed (e.g., battery low detection, temperature threshold). A voltage divider (R_1, R_2) generates a reference voltage V_{ref} at the inverting input.

Reference voltage: $V_{ref} = [R_2 / (R_1 + R_2)] \times VCC$

When $V_{in} > V_{ref} \rightarrow \text{output} = +V_{sat}$. When $V_{in} < V_{ref} \rightarrow \text{output} = -V_{sat}$.

Bypass capacitor C_{BY} across R_2 filters power supply noise. Cutoff frequency: $f_c = 1 / [2\pi(R_1 || R_2)C_{BY}]$

For negative reference: connect $-V_{EE}$ to the voltage divider. Trip point becomes $-V_{ref}$.

2.3 Comparator with Hysteresis – Schmitt Trigger

Problem with simple comparator: noisy signals near the trip point cause rapid switching (chattering/false triggering). Solution: add positive feedback to create TWO trip points – Upper Trip Point (UTP) and Lower Trip Point (LTP).

Schmitt Trigger uses positive feedback (R_1 from output to + input, R_2 from + to ground). The reference voltage at (+) depends on the current output state.

Feedback fraction: $B = R_1 / (R_1 + R_2)$

$UTP = +B \times V_{sat}$ $LTP = -B \times V_{sat}$ Hysteresis $H = UTP - LTP = 2B \times V_{sat}$

Switching sequence: Output at $+V_{sat} \rightarrow (+)$ held at UTP. When V_{in} exceeds UTP, output switches to $-V_{sat}$. (+) now drops to LTP. Output stays at $-V_{sat}$ until V_{in} falls below LTP, then switches back to $+V_{sat}$.

Applications: Noise-immune level detection, waveform shaping (noisy sine $\rightarrow$ clean square), switch debouncing, input buffers in digital logic (74HC14).

3. OSCILLATORS

3.1 Theory of Sinusoidal Oscillation

An oscillator generates a periodic waveform without any external input. It needs an amplifier (gain) and a positive feedback network (frequency selective). The feedback signal replaces the input signal.

Barkhausen Criterion (two conditions must BOTH be satisfied):

1. **Loop Gain = 1: $|A_v \times B| = 1$**
2. **Loop Phase Shift = 0° (or 360°)**

Startup: When power is first applied, loop gain > 1 . Noise voltages are amplified and oscillation builds up. As amplitude grows, gain reduces to exactly 1 and oscillations stabilize at a constant peak-to-peak value.

3.2 Wien-Bridge Oscillator

Most popular audio-frequency oscillator (10 Hz – 100 kHz). Uses a lead-lag circuit (series RC + parallel RC) as the frequency-selective positive feedback network.

Lead-Lag Circuit behavior: At very low freq $\rightarrow$ series C is open, no output. At very high freq $\rightarrow$ shunt C is short, no output. At resonant frequency $f_r \rightarrow$ output is maximum, $B = 1/3$, phase $= 0^\circ$.

Resonant Frequency: $f_r = 1 / (2\pi RC)$

At f_r : Feedback fraction $B = 1/3$, Phase shift $= 0^\circ$

Required amplifier gain: $A_v = 3$ (since $A_v \times B = 3 \times 1/3 = 1$)

Amplitude Stabilization: A tungsten lamp in the negative feedback path acts as automatic gain control (AGC). When amplitude is small $\rightarrow$ lamp is cold $\rightarrow$ resistance low $\rightarrow$ gain $> 3 \rightarrow$ oscillations grow. As amplitude grows $\rightarrow$ lamp heats $\rightarrow$ resistance increases to $R' \rightarrow$ gain drops to exactly 3 $\rightarrow$ stable output.

Closed-loop gain at stability: $A_v(CL) = 2R'/R + 1 = 3$

Two feedback paths: Positive (output $\rightarrow$ lead-lag $\rightarrow$ + input) and Negative (output $\rightarrow$ voltage divider with lamp $\rightarrow$ - input).

3.3 RC Phase Shift Oscillator

Uses only R and C components (no inductors). An inverting amplifier provides 180° phase shift. The feedback network of THREE RC sections provides additional 180° at the oscillation frequency (each $\sim 60^\circ$). Total loop phase $= 360^\circ = 0^\circ$.

Two configurations:

- Three lead circuits (Fig 23-13): Three RC high-pass sections in feedback path.
- Three lag circuits (Fig 23-14): Three RC low-pass sections; amplifier + lag circuits give 360° total.

Oscillation Frequency: $f_r = 1 / (2\pi\sqrt{6 \times RC}) \approx 1 / (15.4RC)$

Minimum amplifier gain required: $|A_v| \geq 29$

Limitation: Not easily tunable. Changing frequency requires changing all three R or C values simultaneously. Used for fixed-frequency, low-power audio applications.

3.4 Colpitts Oscillator

An LC oscillator using a CAPACITIVE voltage divider (C_1 and C_2) in the tank circuit for feedback. Widely used in RF applications. The voltage-divider bias (R_1 , R_2) sets the DC operating point. An RF choke provides high impedance at signal frequency.

Tank circuit: Inductor L with capacitors C_1 and C_2 in series.

Equivalent capacitance: $C = C_1 C_2 / (C_1 + C_2)$

Resonant frequency: $f_r = 1 / (2\pi\sqrt{LC})$

Feedback fraction: $B = C_1/C_2$ Minimum gain: $A_v(\min) = C_2/C_1$

Recognition: Colpitts has capacitive tap (C_1 - C_2 voltage divider). Good frequency stability. Preferred for RF oscillators and VCOs.

3.5 Hartley Oscillator

Similar to Colpitts but uses an INDUCTIVE voltage divider (L_1 and L_2) for feedback. Widely used in AM radio transmitters and receivers.

Tank circuit: Two inductors L_1 , L_2 (or tapped inductor) with capacitor C .

Total inductance: $L = L_1 + L_2$

Resonant frequency: $f_r = 1 / (2\pi\sqrt{LC})$

Feedback fraction: $B = L_2/L_1$ **Minimum gain:** $A_v(\min) = L_1/L_2$

Recognition: Hartley has inductive tap (L_1 - L_2 voltage divider). Easy to tune by varying single capacitor C . Colpitts vs Hartley: same formula structure, just $C \leftrightarrow L$ swapped.

3.6 Crystal Oscillator

When highest frequency accuracy and stability are needed, a quartz crystal oscillator is used. A quartz crystal is a piezoelectric material with a very precise natural resonant frequency.

Equivalent circuit: Series branch (L_s , C_s , R_s) in parallel with plate capacitance C_p . The crystal acts as a LARGE INDUCTOR near its resonant frequency.

Series resonant frequency: $f_r \approx 1 / (2\pi\sqrt{L_s \times C_s})$

Q factor: 10,000 to 1,000,000 (vs LC tank circuit Q of 50–300). Frequency accuracy: $\pm 0.001\%$ or better.

Advantages: Extremely stable, highly accurate, immune to stray capacitances and transistor parameter variations. Applications: microcontroller clocks, GPS, communication systems, RTC modules (32.768 kHz).

4. THE 555 TIMER

The NE555 (also LM555, CA555, MC1455) is the most widely used IC in electronics. It can operate in monostable mode (one stable state) or astable mode (no stable state). Supply: +5V to +15V. Output can source/sink 200mA.

4.1 Internal Structure of 555 Timer

Three 5kΩ resistors in series between VCC and GND create reference voltages:

UTP = $2V_{CC}/3$ (connected to + of upper comparator / threshold comparator)

LTP = $V_{CC}/3$ (connected to - of lower comparator / trigger comparator)

Pin functions: Pin 1–GND, Pin 2–TRIGGER ($< V_{CC}/3 \rightarrow$ output HIGH), Pin 3–OUTPUT, Pin 4–RESET (active LOW, tie to VCC if unused), Pin 5–CONTROL VOLTAGE (bypass with 0.01μF if unused), Pin 6–THRESHOLD ($> 2V_{CC}/3 \rightarrow$ output LOW), Pin 7–DISCHARGE (discharges external capacitor), Pin 8–VCC.

4.2 Monostable Operation (One-Shot)

ONE stable state: output LOW. A negative trigger pulse on pin 2 causes the output to go HIGH for a fixed time W, then return to LOW automatically. External components: R between VCC and pin 7, C between pin 6/7 and GND.

Step-by-step operation:

1. Stable state: Q=HIGH $\rightarrow$ discharge transistor ON $\rightarrow$ C clamped at 0V $\rightarrow$ output LOW.
2. Trigger: Pin 2 drops below $V_{CC}/3 \rightarrow$ flip-flop sets $\rightarrow$ Q=LOW $\rightarrow$ transistor OFF $\rightarrow$ C charges through R.
3. C charges exponentially toward VCC (time constant = RC).
4. When $V_c > 2V_{CC}/3$: upper comparator resets flip-flop $\rightarrow$ Q=HIGH $\rightarrow$ transistor ON $\rightarrow$ C discharges instantly $\rightarrow$ output LOW.

Pulse Width: $W = 1.1 \times R \times C$

Applications: Pulse generation, time delays, switch debouncing, pulse stretcher, missing pulse detector, frequency divider.

4.3 Astable Operation (Free-Running Oscillator)

NO stable state. The output continuously oscillates between HIGH and LOW producing a rectangular wave. External components: R1 (VCC to pin 7), R2 (pin 7 to pins 2/6), C (pins 2/6 to GND). Pins 2 and 6 are shorted together.

Step-by-step operation:

1. C charges through (R1 + R2) toward VCC. Output is HIGH.
2. When V_c reaches $2V_{CC}/3$: flip-flop resets $\rightarrow$ output LOW $\rightarrow$ transistor ON (pin 7 grounded).
3. C discharges through R2 only (not through R1). Output stays LOW.
4. When V_c drops to $V_{CC}/3$: flip-flop sets $\rightarrow$ output HIGH $\rightarrow$ transistor OFF. Cycle repeats.

HIGH time: $W = 0.693 \times (R1 + R2) \times C$

Period: $T = 0.693 \times (R1 + 2R2) \times C$

Frequency: $f = 1.44 / [(R1 + 2R2) \times C]$

Duty cycle: $D = (R1 + R2) / (R1 + 2R2)$ [always > 50%]

Note: Duty cycle > 50% because C charges through R1+R2 but discharges through R2 only. For exactly 50%: add diode in parallel with R2.

Applications: Clock generator, LED blinker, tone generator, PWM, motor speed control, IR transmitter, VCO (vary pin 5 voltage to change frequency).

5. QUICK REVISION — ALL FORMULAS & KEY POINTS

★ USE THIS SECTION BEFORE VTU EXAMS ★

5.1 All Key Formulas

LINEAR OP-AMP CIRCUITS

Summing amplifier:	$V_{out} = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$
Averager (n inputs):	$V_{out} = -(V_1+V_2+\dots+V_n)/n$ [$R_{in}=nR$, $R_f=R$]
R/2R DAC:	$BIN = D_0+2D_1+4D_2+8D_3$; $V_{out} =$
$-(BIN/2^N) \times 2V_{ref}$	

COMPARATORS

Min input to saturate:	$V_{in(min)} = \pm V_{sat}/A_{VOL}$
Non-zero reference:	$V_{ref} = R_2/(R_1+R_2) \times V_{CC}$
Bypass cutoff freq:	$f_c = 1/[2\pi(R_1 \parallel R_2)C_{BY}]$
Schmitt feedback:	$B = R_1/(R_1+R_2)$
Trip points:	$UTP = +B \times V_{sat}$ $LTP = -B \times V_{sat}$
Hysteresis:	$H = UTP - LTP = 2B \times V_{sat}$

OSCILLATORS

Wien-Bridge:	$f_r = 1/(2\pi RC)$; $B=1/3$; $A_v=3$
RC Phase Shift:	$f_r = 1/(2\pi\sqrt{6} \cdot RC) \approx 1/(15.4RC)$; $A_v(min)=29$
Colpitts: $C=C_1C_2/(C_1+C_2)$;	$f_r=1/(2\pi\sqrt{LC})$; $B=C_1/C_2$; $A_v(min)=C_2/C_1$
Hartley: $L=L_1+L_2$;	$f_r=1/(2\pi\sqrt{LC})$; $B=L_2/L_1$; $A_v(min)=L_1/L_2$
Crystal:	$f_r \approx 1/(2\pi\sqrt{L_sC_s})$; $Q = 10,000$ to $1,000,000$

555 TIMER

Monostable:	$W = 1.1RC$ (pulse width)
Astable HIGH time:	$W = 0.693(R_1+R_2)C$
Astable period:	$T = 0.693(R_1+2R_2)C$
Astable frequency:	$f = 1.44/[(R_1+2R_2)C]$
Duty cycle:	$D = (R_1+R_2)/(R_1+2R_2)$ [always > 50%]
Trip points:	$UTP = 2V_{CC}/3$ $LTP = V_{CC}/3$

5.2 Key Concept Reminders

Virtual ground: Inverting input is at 0V in negative feedback op-amp circuits.

Barkhausen: BOTH conditions must hold $\rightarrow |A_vB|=1$ AND loop phase=0°.

Colpitts = Capacitors (C1-C2 divider). Hartley = H = Henry = Inductors (L1-L2 divider).

Wien-Bridge: Remember the 3s $\rightarrow B=1/3$, $A_v=3$, $f_r=1/(2\pi RC)$.

RC Phase Shift: 3 stages $\times 60^\circ = 180^\circ$. $A_v \geq 29$.

Schmitt: Higher B = more hysteresis = more noise immunity but wider dead zone.

555 astable: C charges through R_1+R_2 , discharges through R_2 only $\rightarrow$ duty cycle always > 50%.

R/2R ladder advantage: only 2 resistor values needed for any number of bits.

Crystal Q factor = 10,000 to 1,000,000 vs LC tank Q of 50–300. Best stability.

5.3 VTU Important Questions

1. Derive the output voltage expression for an inverting summing amplifier with n inputs.
2. Explain the 4-bit binary-weighted D/A converter with circuit diagram. Show output waveform.
3. Explain R/2R ladder D/A converter. Derive the output voltage formula. Solved example.
4. Explain non-inverting and inverting comparators with zero reference. Draw waveforms.
5. What is hysteresis? Derive UTP, LTP and H for an inverting Schmitt trigger.
6. State and explain the Barkhausen criterion for sustained oscillation.
7. Explain Wien-Bridge oscillator with circuit diagram. Derive f_r . Explain AGC using tungsten lamp.
8. Explain RC phase shift oscillator. Derive f_r . What is the minimum gain required?
9. Compare Colpitts and Hartley oscillators with circuit diagrams and formulas.
10. Explain monostable operation of 555 timer. Derive $W = 1.1RC$.
11. Explain astable operation of 555 timer. Derive T, f, and duty cycle D.
12. Why is crystal oscillator preferred over LC? Explain with circuit diagram.

— *End of Module 3 Notes* — vtU-easy.com —